University of BRISTOL

AVDASI-3: Sustainable Aviation & Ethics

Lecture-8: Recent trends in UAM / AAM industry

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Urban air mobility (UAM)

Small objects fly better

- Lift $\propto x^2$ & Weight $\propto x^3$
- Low Re $\Rightarrow$ Laminar $\Rightarrow$ Less drag

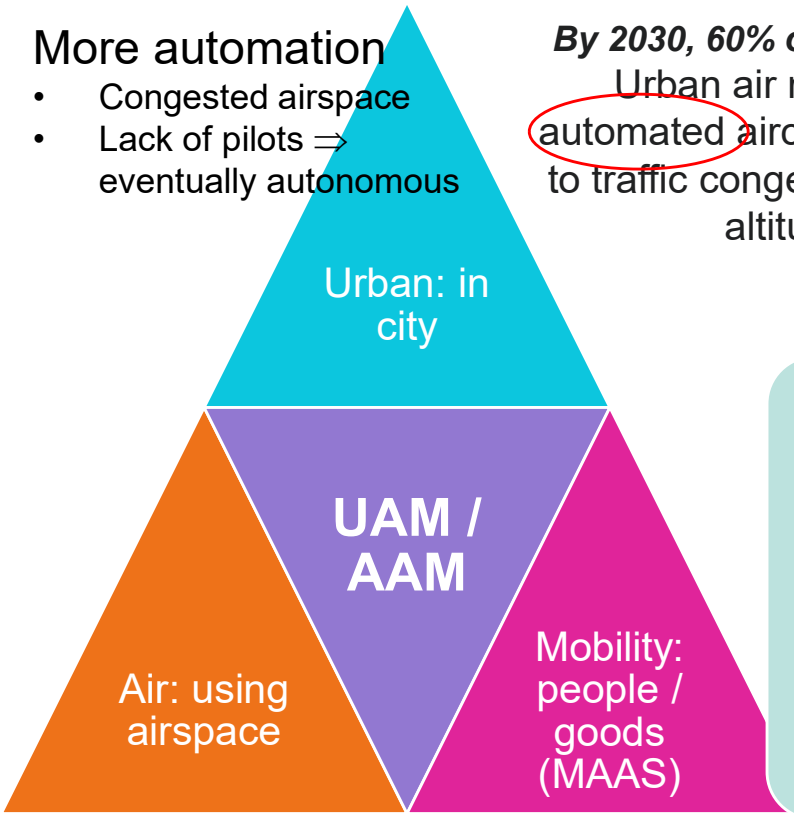
Demand driven

More automation

- Congested airspace
- Lack of pilots $\Rightarrow$ eventually autonomous

By 2030, 60% of world population of 5bn will be living in city

Urban air mobility (UAM) is the use of small, highly automated aircraft which have been developed in response to traffic congestion, to carry passengers or cargo at lower altitudes in urban and suburban areas

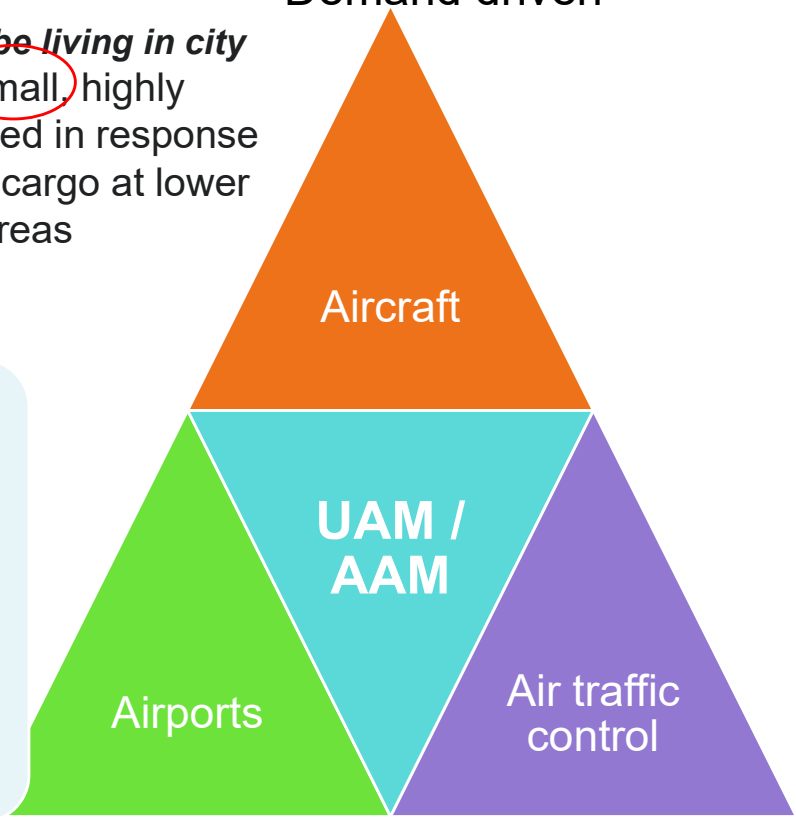


Reach

- Intracity (airport)
- Intercity (rural)
- Nearby islands / mountains / remote area etc.

Purpose

- Business commute
- Leisure / tourism
- Medical emergency



Market perspective

Demand pull

- Traffic congestion
- Air pollution
- Noise pollution
- Safety

Technology push

- Newer batteries
- Novel D&A sensors and CLAW
- Better communication
- Higher computation power

Other Challenges

- Infra (vertiport, MRO)
- Last mile (MAAS)
- Airspace management & certification
- Political & social acceptance

New aircraft

- New configuration✓
- Distributed electric propulsion✓
- New structural concepts✓
- Advanced avionics & sensors
- Autonomy, AI etc.

New business model

- Ground infrastructure✓
- Connected pax (Apps)
- Competitive & dynamic pricing
- Cab aggregators for last mile?

The past & the future

LA Airways – Heli services from LA airport to downtown LA, Hollywood, Disneyland etc.

VOOM (Airbus sub.) on-demand Heli aggregator @ Sau Paolo, Mexico City & SFO – closed during COVID but harbinger of UAM industry

City AB-NG

EHang

Volocopter

Lilium

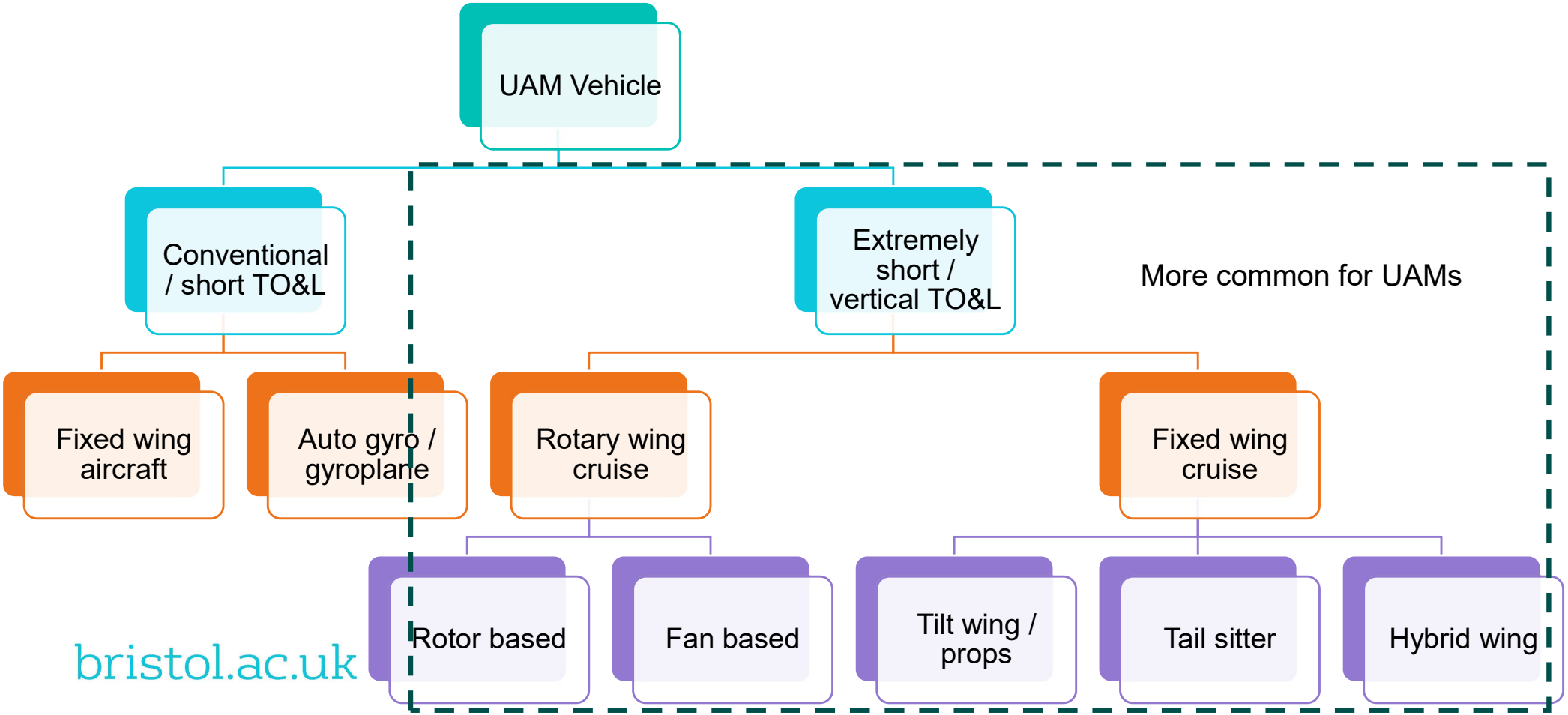
Joby

Vertical

Archer

1950-60s → 1970-80s → 2010-20s → Niche market to start with, then democratisation… → Piloted to start with, followed by OPV & then autonomous

UAM vehicle architecture



Vertidrome

- Need for novel aerodrome
  - Airport: High performance, cost optimised for FW
  - Heliport: Low capacity, more expensive
  - UAM / Vertiport (~ 20m x 20m per pad)
    - On demand high density, but affordable
    - Surrounding obstacles inside city limits, i.e. steep approaches / departures
    - Autonomous ops
    - New aircraft configurations / technologies compatible

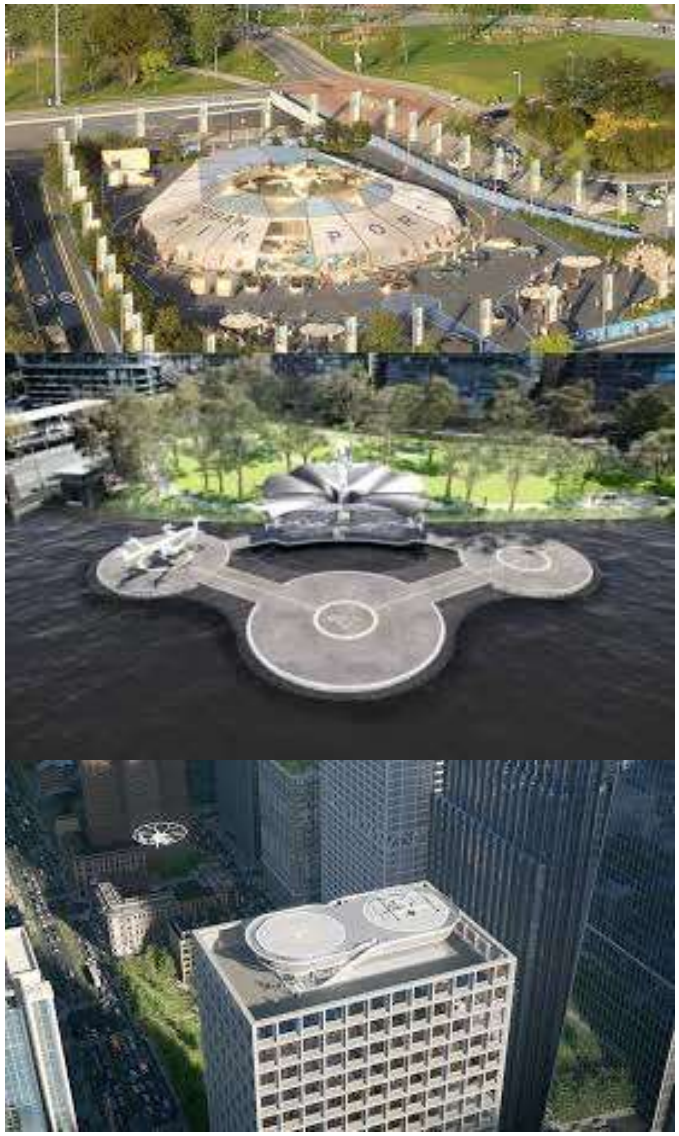
Vertidromes

- Types
  - Vertistop (suburban): Small (1/2 pads), only boarding / un boarding, no charging, no MRO
  - Vertiport (downtown): Large (multiple pads), charging, light MRO etc.
  - Vertihub (outskirts): Very large, power grid connected, heavy MRO
- Typical performance matrices
  - Pax delay < 2 min
  - Operator delay < 5 min
  - In time flights > 95% etc.

Microsoft Word - NASA Vertiport Trade Study 20210119_FINAL (FOR DELIVERY).docx

Vertidrome challenges

- Space in dense cities (ground / floating / rooftops etc.)
- Arrival / departure procedures
- Scheduling & spacing
- Emergency procedures
- Ground handling
- Passenger flow handling
- Public acceptance of **noise levels**



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Effective Perceived Noise Level

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Sound Pressure Levels (SPL)

The human ear is extremely sensitive to fluctuations in acoustic pressure

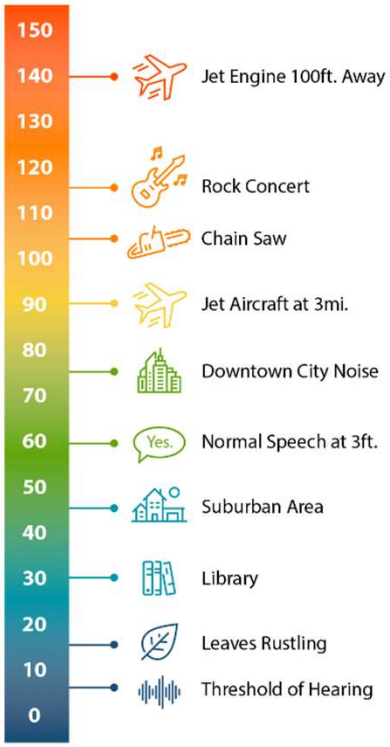
This is why we need a **decibel (dB) scale**

$$\text{SPL (dB)} = 10\log_{10}(p^2/p_{\text{ref}}^2)$$

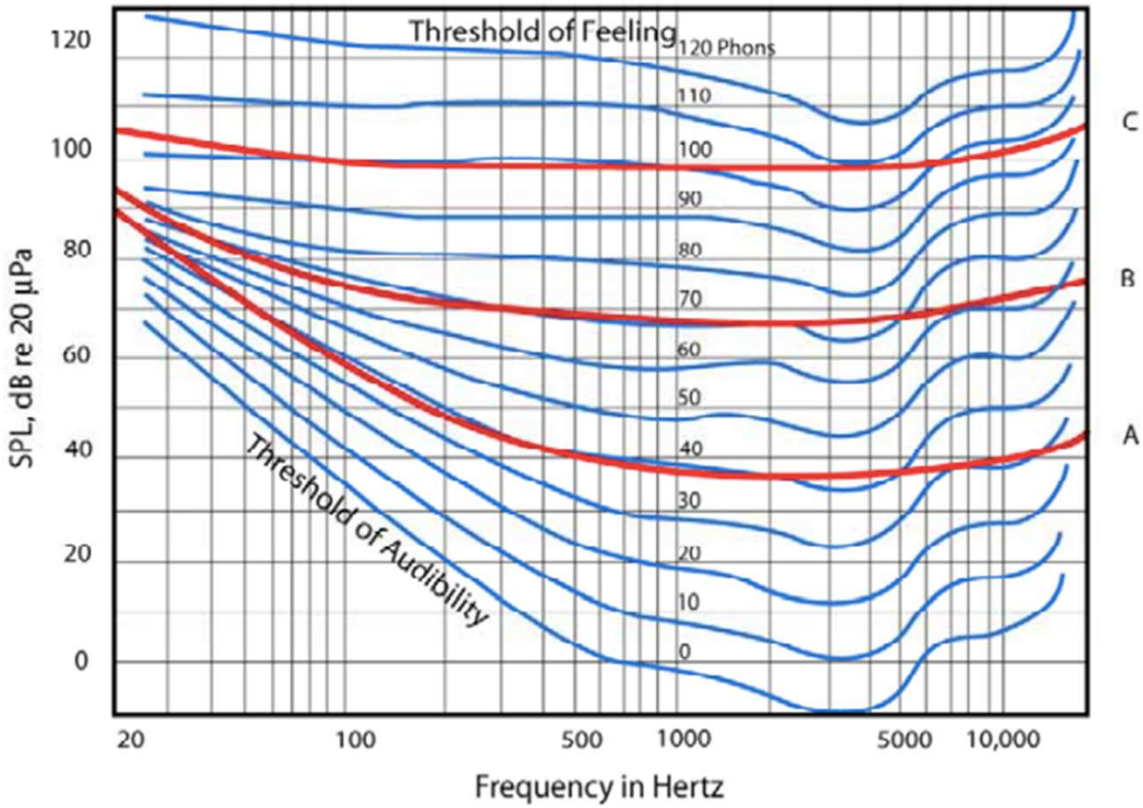
- 0 dB = $p_{\text{ref}} = 20 \times 10^{-6}$ Pa
- 60 dB = 20×10^{-3} Pa (1000 × greater pressure than 0 dB)
- 120 dB = 20 Pa (1000 × greater pressure than 60 dB)

6 dB = half/double the acoustic pressure

$$\text{SPL (dB) variation with distance: } L_2 = L_1 - 20 * \log\left(\frac{r_2}{r_1}\right)$$



Threshold of feeling



Fletcher-Munson Curves
(John Hopkins University)

Effective Perceived Noise Levels (EPNL)

- EPNdB (flyover) $= (0.0125T.N^2 + 104.6) - (40.25N + 2.9)(T.N/mg)$
- EPNdB (lateral) $= 7.67\log_{10}(T.N) + 76.44$
- EPNdB (approach) $= 0.5486x + 101.18$

Where $x = 10\log_{10}[\, M_0^6(FF\sin^2(\delta F) + k\Sigma(n.d^2))\,] + 10\log_{10}(0.34T.N/D_f^2) + 10\log_{10}(D_f^2) + 10\log_{10}(N)$

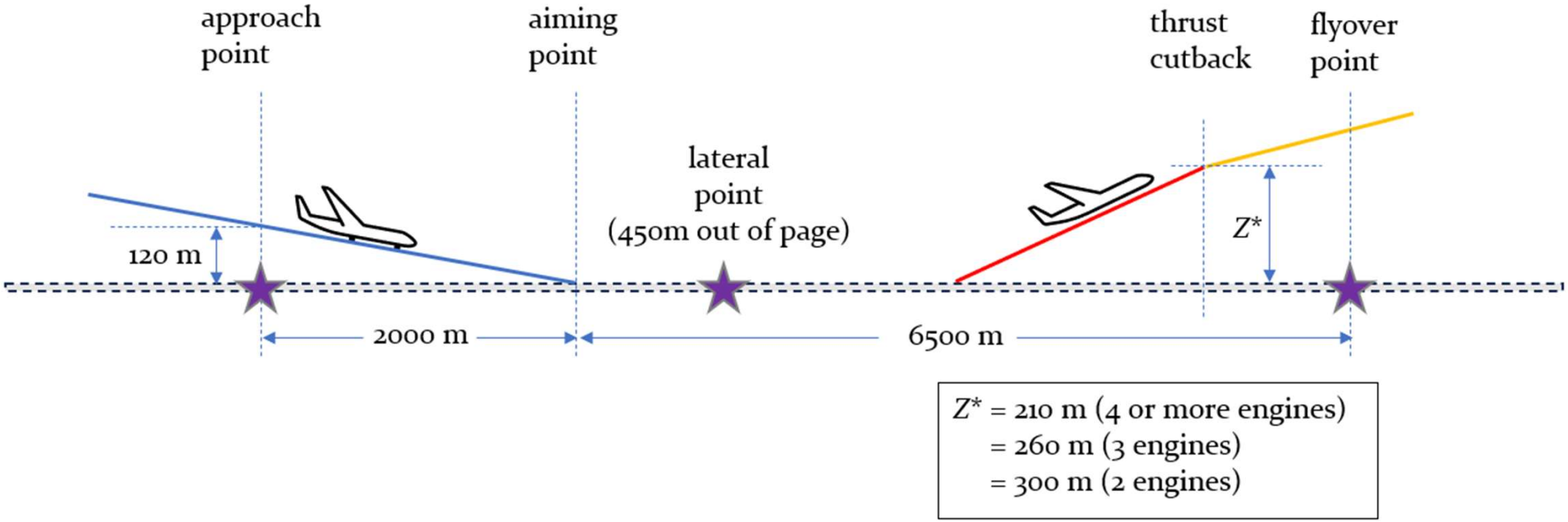
N =	Number of engines
T =	Thrust of one engine (kN)
m =	Aircraft mass (tonnes) [MTOW], and g = 9.81 m/s^2
M_0 =	Mach number $=1.3V_s/c_0$: V_s = approach speed (m/s), c_0 = 343 m/s
FF =	S_F/S, where S_F = flap area and S = wing area
δ_F =	Flap angle (degrees) on approach (usually less than landing)
k =	constant depending on landing configuration: • 4 wheels per leg → k =1.22 • 2 wheels per leg → k =1.56
n =	Number of wheels of diameter, d (m)
D_f =	Fan diameter (m)

Equations are statistical fit, interpolate at your own peril

Key points:

- Lateral measurements are solely engine jet noise
- Aircraft mass influences flyover noise
- Approach is much more complex, involving wing, landing gear and engine noise

International Civil Aviation Organization (ICAO) noise measurement positions



ICAO noise requirements

Maximum allowable EPNdB (EPNdB = Effective Perceived Noise in dB)

M = Maximum take-off mass in 1 000 kg		0 – 20.2	20.2 – 28.6	28.6 – 35	35 – 48.1	48.1 – 280	280 – 385	385 – 400
Lateral full-power noise level (EPNdB) All aeroplanes		94			80.87 + 8.51 log M			103
Approach noise level (EPNdB) All aeroplanes		98			86.03 + 7.75 log M		105	
Flyover noise levels (EPNdB)	2 engines or less	89			66.65 + 13.29 log M			101
	3 engines	89		69.65 + 13.29 log M				104
	4 engines or more	89	71.65 + 13.29 log M					106

https://www.icao.int/environmental-protection/pages/reduction-of-noise-at-source.aspx

Example calculation

Ne	Number of engines	[2]
T	Thrust (kN) of one engine, usually given in lbf units	[23000lb]
W	Aircraft Mass (tonnes)	[MTOW = 66tonnes]
M	Mach Number take as 1.3Vs/a	
	a is the speed of sound at sea-level [340.29m/s]	
	Vs = 53.86m/s at MLW with δ_f=20° (approach)	
	Vs = 51.40m/s at MLW with δ_f=40° (landing)	
FF	Flap area as a fraction of wing area	(S_F/S)
S_F	Flap area (m^2)	[21.1m^2]
S	Wing area (m^2)	[122.4m^2]
δ_F	Flap angle (deg) on approach (usually less than landing)	[20°]
k	constant depending on landing configuration	
	4 wheels per leg	k=1.22
	2 wheels per leg	k=1.56
n	Number of wheels of diameter d (m)	
	MLG 2 wheels/leg	[d=49ins]
	NLG 2 wheels	[d=30ins]
D_f	Fan diameter (m)	usually given in inches [68.3ins]
g	Acceleration due to gravity (9.81m/sec^2)	

Flyover Calculation

$$\begin{aligned}
EPNdB(flyover) &= ((0.0125*2*2*102.31)+104.6) \\
&\quad -[(40.25*2+2.9)*((2*102.31)/(66*9.81)) \\
&= 5.1155+104.6-(83.4*0.316) \\
&= \mathbf{83.34} \text{ (ICAO target is 90.83)}
\end{aligned}$$

Sideline Calculation

$$\begin{aligned}
EPNdB(sideline) &= 7.67\ Log_{10}\ (2*102.31) + 76.44 \\
&= (7.67 * 2.3109) + 76.44 \\
&= 17.72 + 76.44 \\
&= \mathbf{94.16} \text{ (ICAO target is 96.35)}
\end{aligned}$$

Approach Calculation

$$\begin{aligned}
M &= 1.3*53.86/340.29 = 0.2058 \\
FF*\sin^2(\delta_F) &= (21.1/122.4) * \sin^2(20) \\
&= 0.1724 * 0.117 = 0.0202 \\
\Sigma knd^2 &= ((4*1.2446^2 + 2*0.762^2) * 1.56) \\
&= 11.478 \\
x &= 10Log_{10}(0.2058^6*(0.0202+11.478)) \\
&\quad + 10Log_{10}(2*0.34*102.31/1.735^2) \\
&\quad + 20Log_{10}(1.735) + 10Log_{10}(2) \\
&= -30.587 + 13.638 + 4.786 + 3.01 \\
&= -9.153 \\
EPNdB(approach) &= 0.5486 * (-9.153) + 101.18 \\
&= \mathbf{96.16} \text{ (ICAO target is 100.13)}
\end{aligned}$$

L_{den} metric (Europe)

- Lden = Averaged noise (L) of all single point noise events over day, evening & night (den) across the year
- Various definitions & equations used across different geographies and with different assumptions / simplifications
  - Day event (0700-1900 hrs): no penalty
  - Evening event (1900-2300 hrs): 5 dB penalty (not used in USA)
  - Night event (2300-0700 hrs): 10 dB penalty
- Calculations

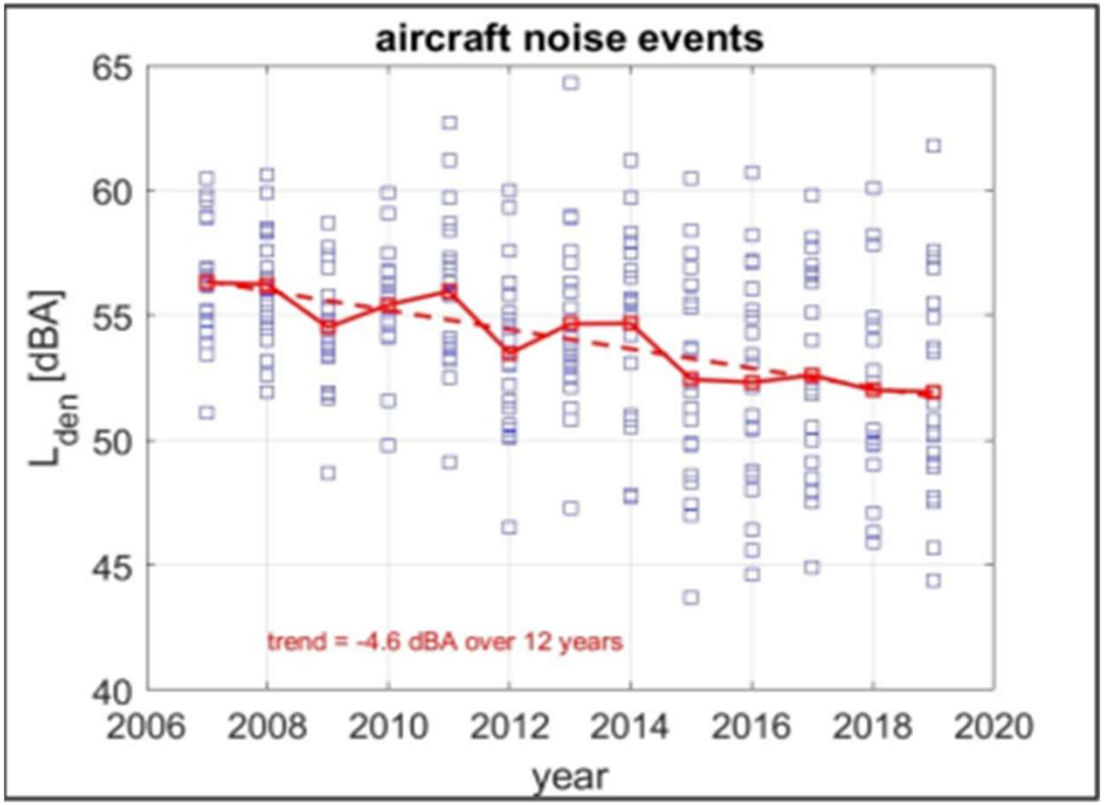
$$\mathbf{L_{den} = 10\log\Delta t + L + 10\log(N_d + 3.16N_e + 10N_n) - 75}$$

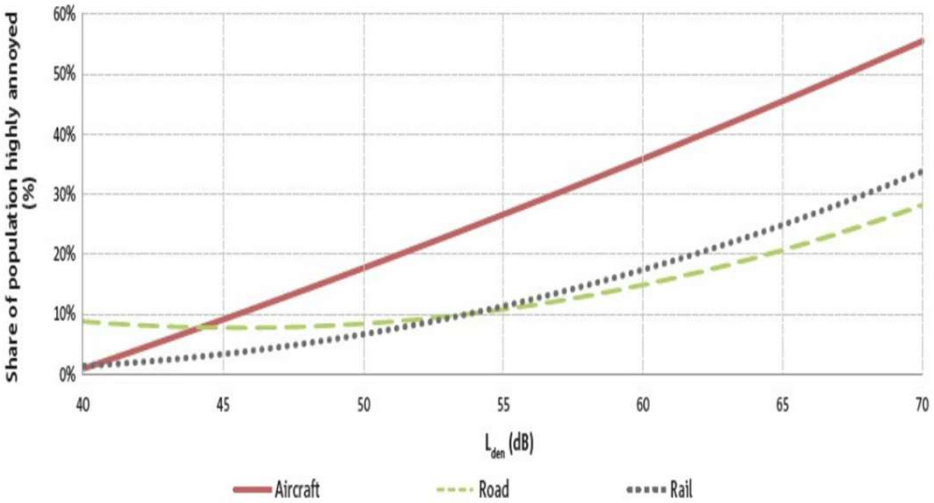
where Δt = time duration for each noise event (s)

L = noise per aircraft (dB)

N_d / N_e / N_n = no. of flights per year during day / evening / night

L_{den} benchmarks





L_{den} = 58dB is often considered a threshold wherein about 30% of the populations is highly annoyed

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Thank you

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